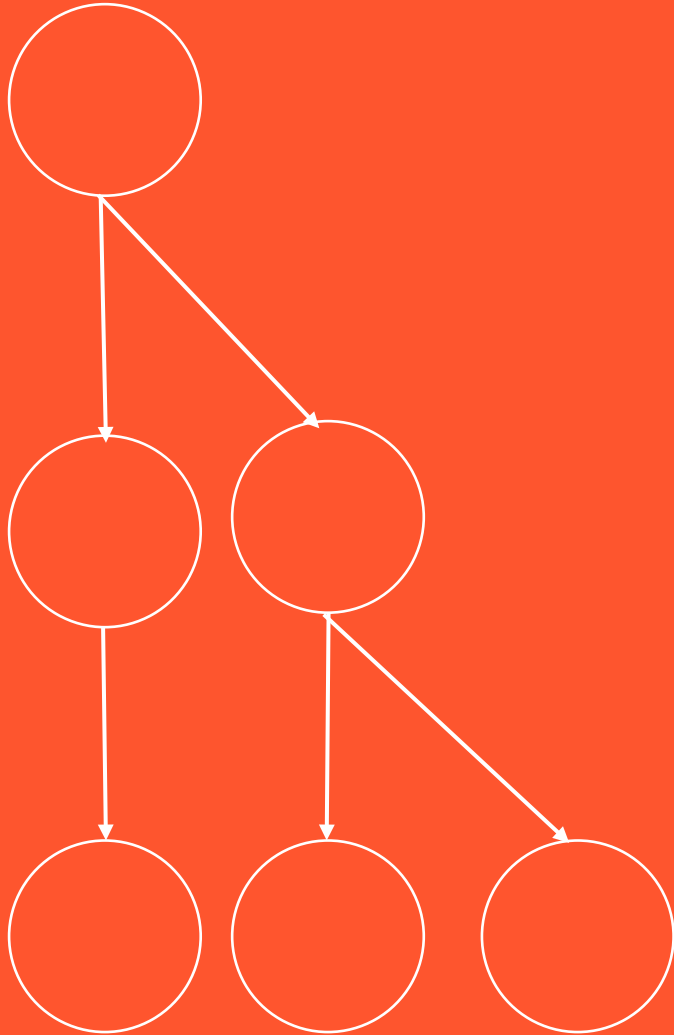


BCOG 100

Symbolic Cognition

Lecture 1: Symbolic Cognition and the Cultural Explosion



UNIVERSITY OF
ILLINOIS
URBANA-CHAMPAIGN

Reading Check 12-1

Password: symbol

Question: What is the Physical Symbol Systems hypothesis?

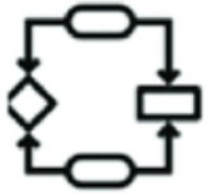


Symbolic Cognition and the Cultural Explosion



Computation

What does the system do?



Algorithm

How does the system do it?



Implementation

How is the system realized?

Recap & Overview

Last Time:

Language in the Mind and Brain

Today:

This week: evolution of symbolic cognition and the human cultural explosion

Symbolic Cognition and the Cultural Explosion

SEMIOTICS

Semiotics is the study of signs. It is both the theory & analysis of signs & signifying practices.



Ferdinand de Saussure
(1857-1913)

Saussure was a french linguist that was one of the first to produce a semiotic theory during work in the early 1900's. Saussure stated that a sign is made up of two parts, the SIGNIFIER and the SIGNIFIED.

Charles Sandes Peirce, was an American philosopher that also formulated a semiotic theory around the same time as Saussure, the early 1900's. Pierce stated semiotics to be the relationship between a SYMBOL, ICON and an INDEX.



Charles Sanders Peirce
(1839-1914)

SIGN

A successful, understandable form of communication.

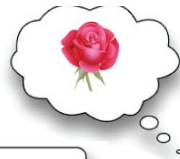
SIGNIFIER

Any motion, gesture, image, sound, pattern, or event that conveys meaning and communicates.



CONNOTATION

The feelings, ideas and cultural meanings which are associated with the word or object. Not the literal meaning.



Rose

SIGNIFIED

The concept that a signifier refers to, the meaning it conveys.



DENOTATION

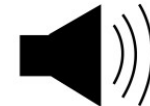
The literal or primary meaning of a word, in contrast to the feelings or ideas that the word suggests.

What Is a Symbol?

- Icons, indices, and symbols
- Arbitrary form, shared meaning
- The power of representation

ICONS

Signs where the signifier resembles the signified.



SYMBOLS

Signs where the relation between signifier and signified is purely conventional and culturally specific.



INDEXES

Signs where the signifier is caused by the signified.



Syntagm

A single word or element of design that, when altered, can change the overall meaning.



Paradigm

A group or related ideas that are interchangeable in a sentence or design, yet they still have the same collective meaning.

Symbolic Cognition and the Cultural Explosion



Why Symbols Matter

- Tools for shared imagination
- Bridges between minds and time
- The engine of cumulative culture

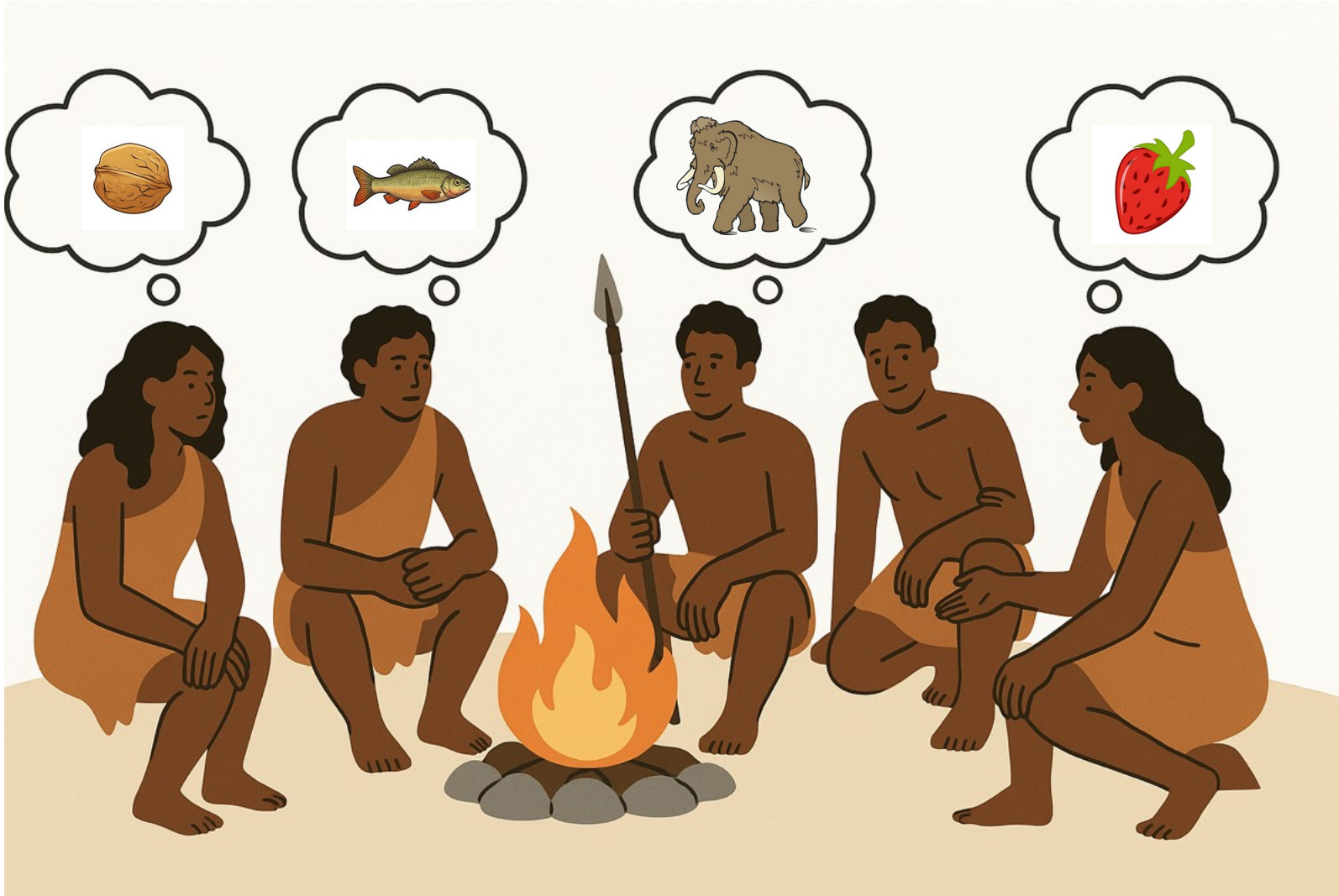
tools

speech
symbols

writing

science

Symbolic Cognition and the Cultural Explosion



The Functional Problem

- Why did symbols evolve?
- What problem did they solve?
- Coordination, cooperation, and memory

Symbolic Cognition and the Cultural Explosion

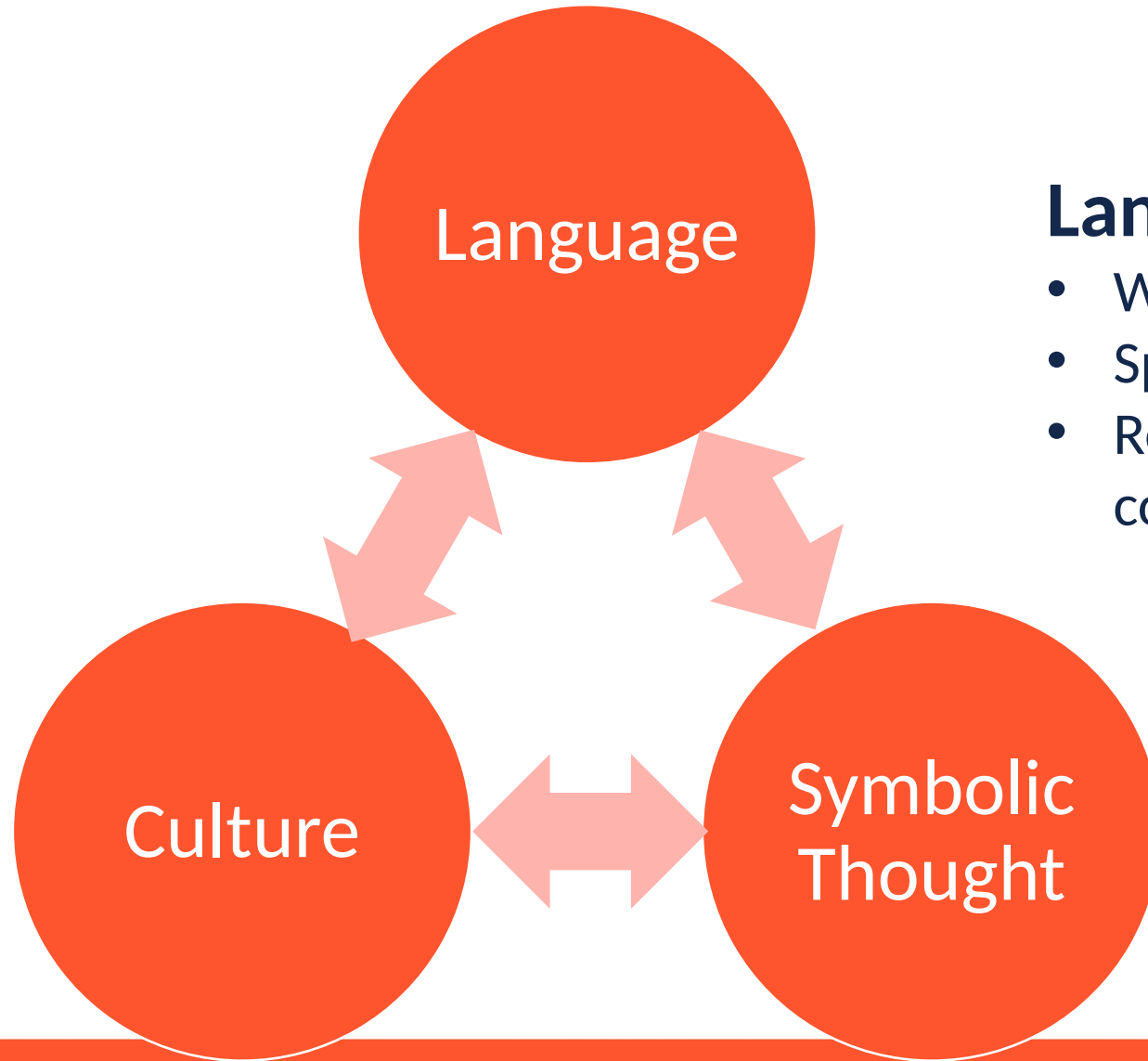
The Evolutionary Threshold

- A sudden flowering of culture
- Art, ornament, and ritual appear
- The dawn of symbolic behavior

Years Ago	Species / Site / Period	Cultural or Cognitive Development
3.3–2.6 million	<i>Australopithecus</i> , early <i>Homo</i>	Earliest stone tools
1.7–1.4 million	<i>Homo erectus</i>	Acheulean hand axes; symmetry and standardized design—evidence of planning and teaching.
800,000–500,000	<i>Homo heidelbergensis</i> and <i>Homo erectus</i>	Fire control, coordinated hunting, possible proto-language; increased cooperation.
400,000–300,000	Early <i>Homo sapiens</i> and Neanderthals	Wooden spears, pigment use (red ochre); possible early symbolic marking.
250,000–160,000	<i>Homo sapiens</i> (Africa)	Long-distance raw material transport, shell ornaments; beginnings of trade and identity signaling.
100,000–75,000	Blombos Cave (South Africa)	Engraving, perforated shells; clear evidence of symbolism.
70,000–50,000	“Cognitive Revolution”	Rapid innovation in tools, art, and social networks.
50,000–35,000	Upper Paleolithic (Europe)	Cave art (Lascaux, Chauvet), figurines (<i>Venus of Willendorf</i>), bone flutes; full symbolic expression.
30,000–10,000	Late Paleolithic to Neolithic	Complex burials, long-distance trade, ritual structures (Göbekli Tepe); domestication begins.
10,000–5,000	Agricultural Revolution	Permanent settlements, writing precursors (tokens, tallies), calendrical systems.
5,000–3,000	Early Civilizations (Sumer, Egypt, Indus)	Writing systems (cuneiform, hieroglyphs), organized religion, codified law.



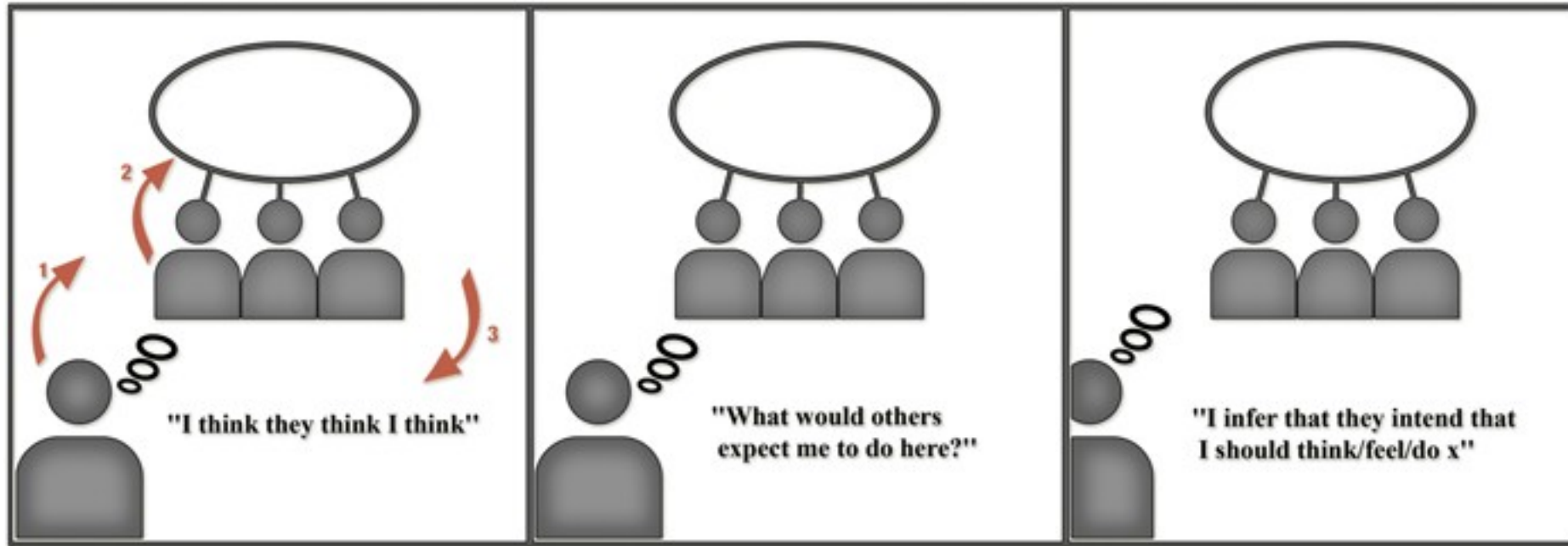
Symbolic Cognition and the Cultural Explosion



Language and Symbolic Thought

- Which came first—or did they co-evolve?
- Speech and symbol shaped each other
- Representation bootstrapping communication

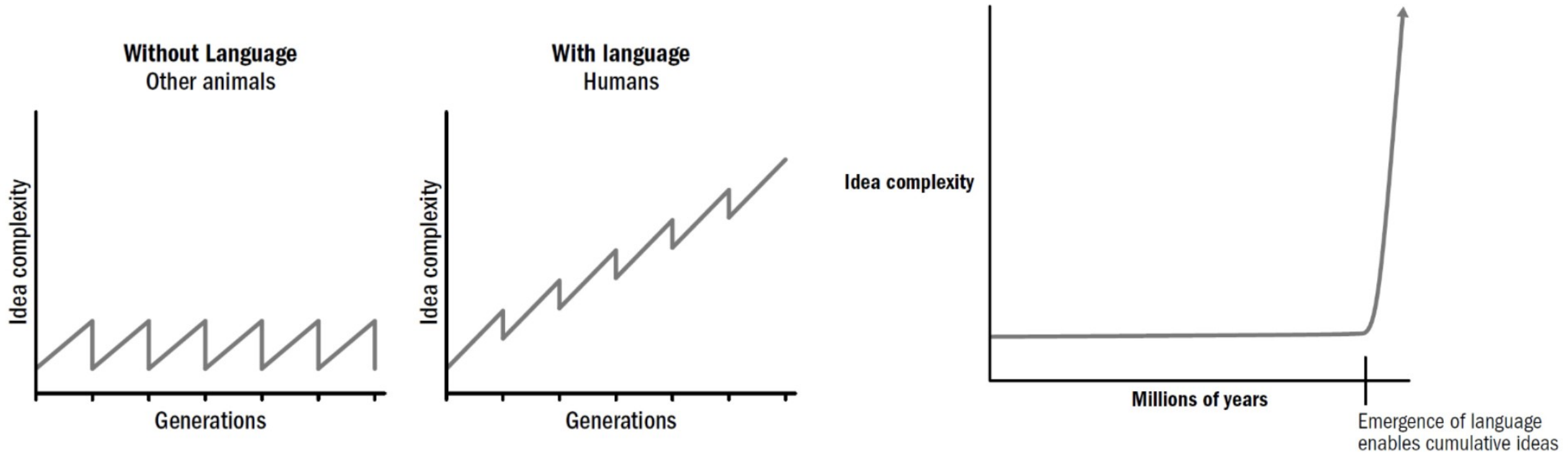
Symbolic Cognition and the Cultural Explosion



Shared Intentionality

- Meaning through mutual understanding
- Thinking about minds together
- The social foundation of symbols

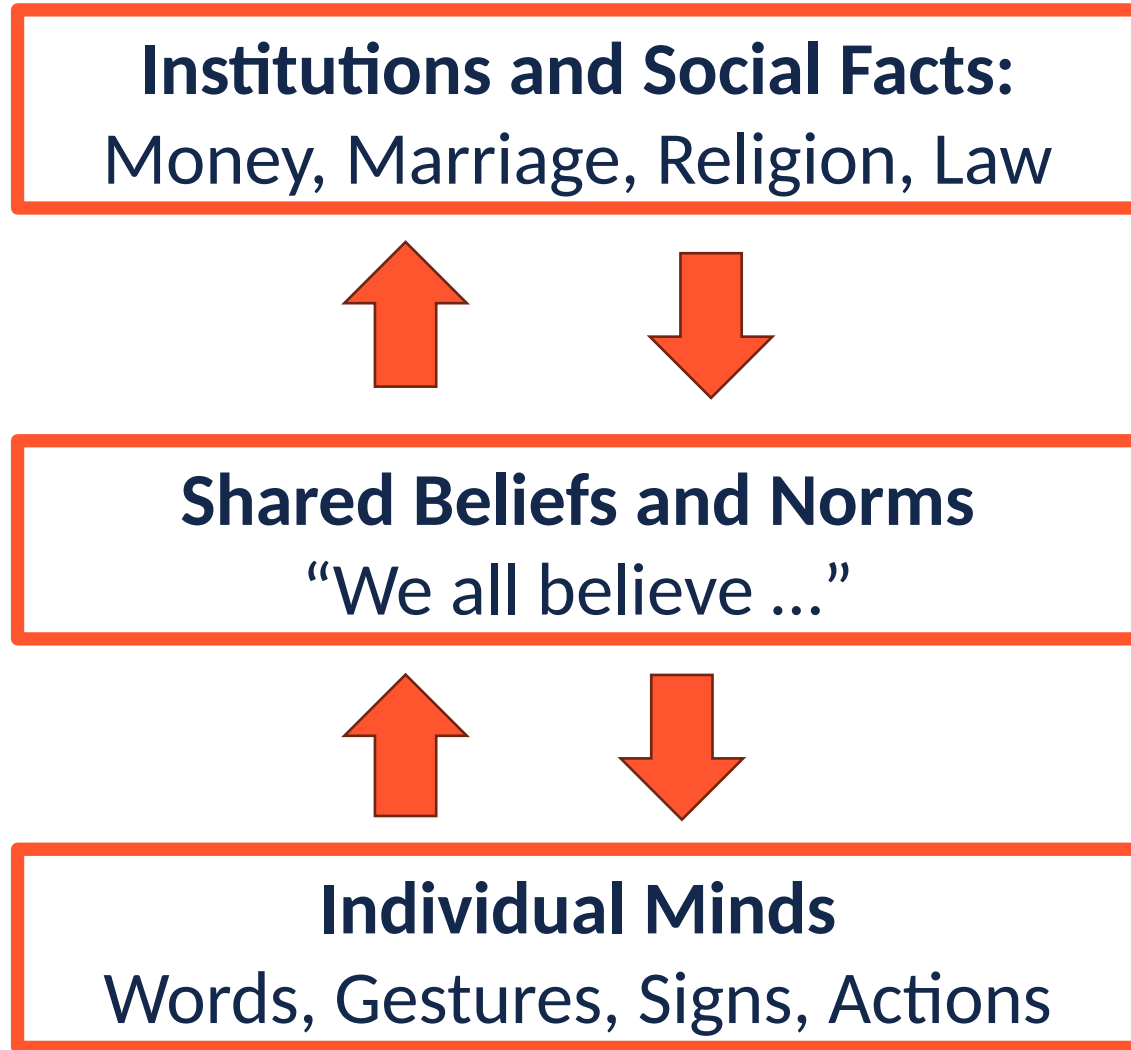
Symbolic Cognition and the Cultural Explosion



Teaching, Imitation, and Cumulative Culture

- Learning from others
- Building on what came before
- Culture as the second inheritance system

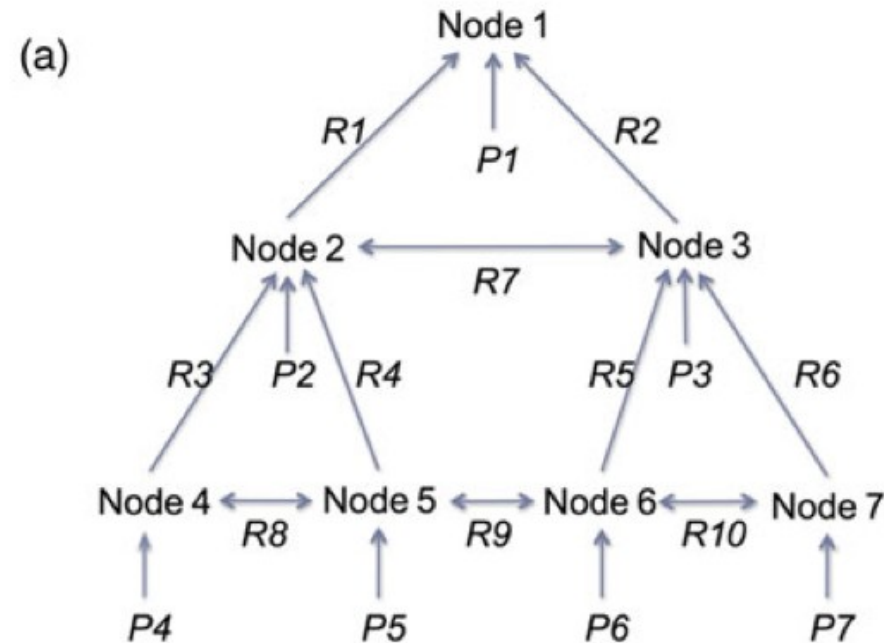
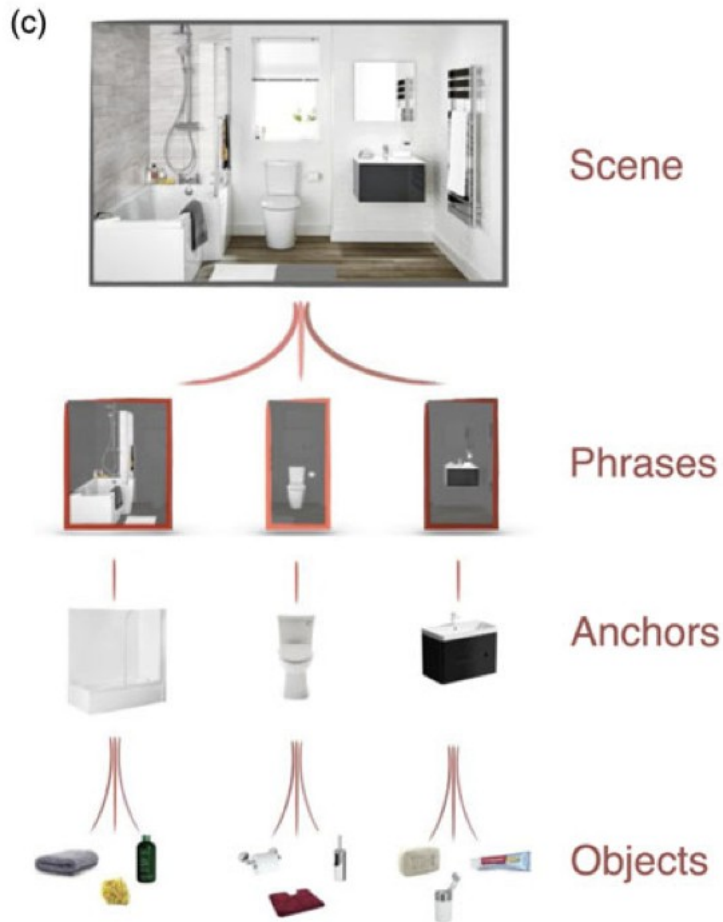
Symbolic Cognition and the Cultural Explosion



Collective Intentionality and Institutions

- Shared beliefs about shared beliefs
- Symbols that organize societies
- The invention of the social world

Symbolic Cognition and the Cultural Explosion

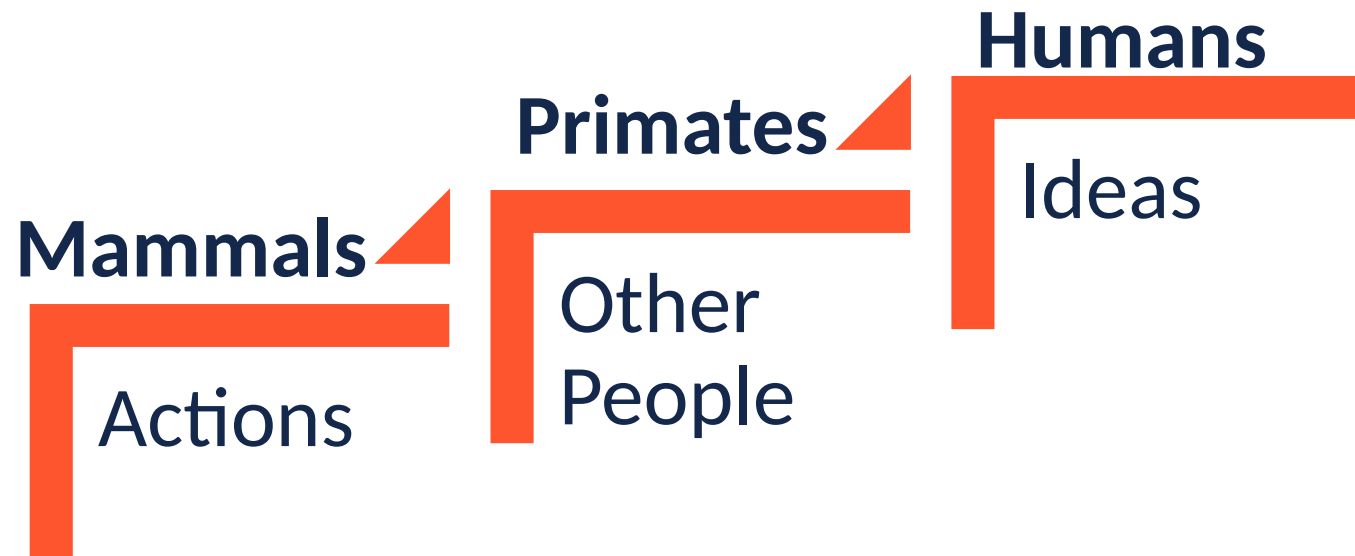


The Cognitive Architecture of Symbolic Thought

- Representations that combine
- Rules that generate meaning
- Structure at the core of thought

Symbolic Cognition and the Cultural Explosion

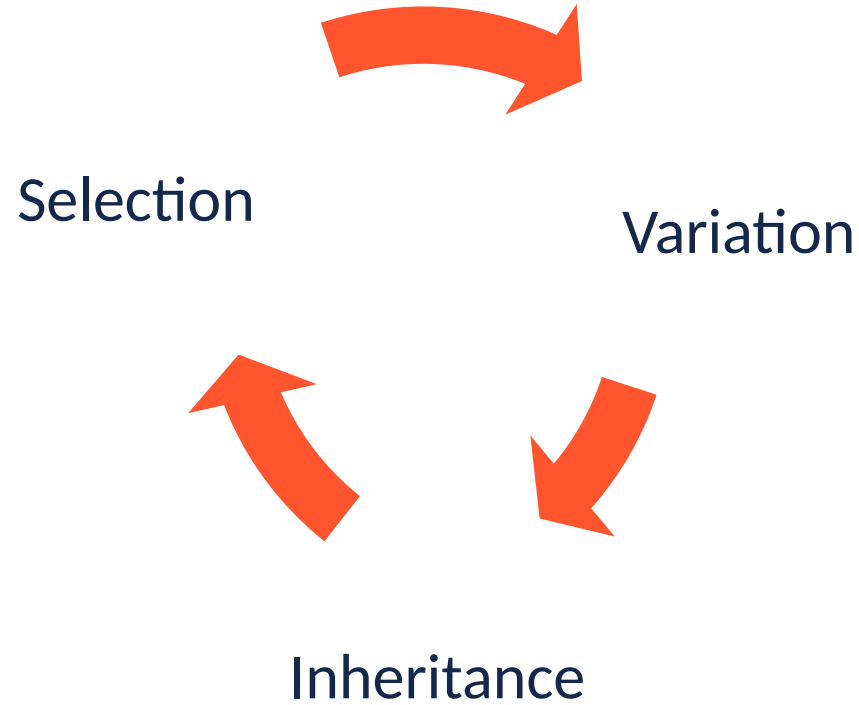
Simulation Staircase



Symbols and Simulation

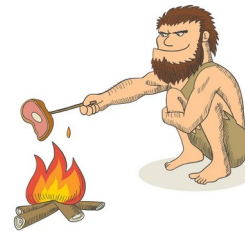
- Imagination made explicit
- From vicarious learning to symbolic modeling
- Thinking through representations of thought

Symbolic Cognition and the Cultural Explosion

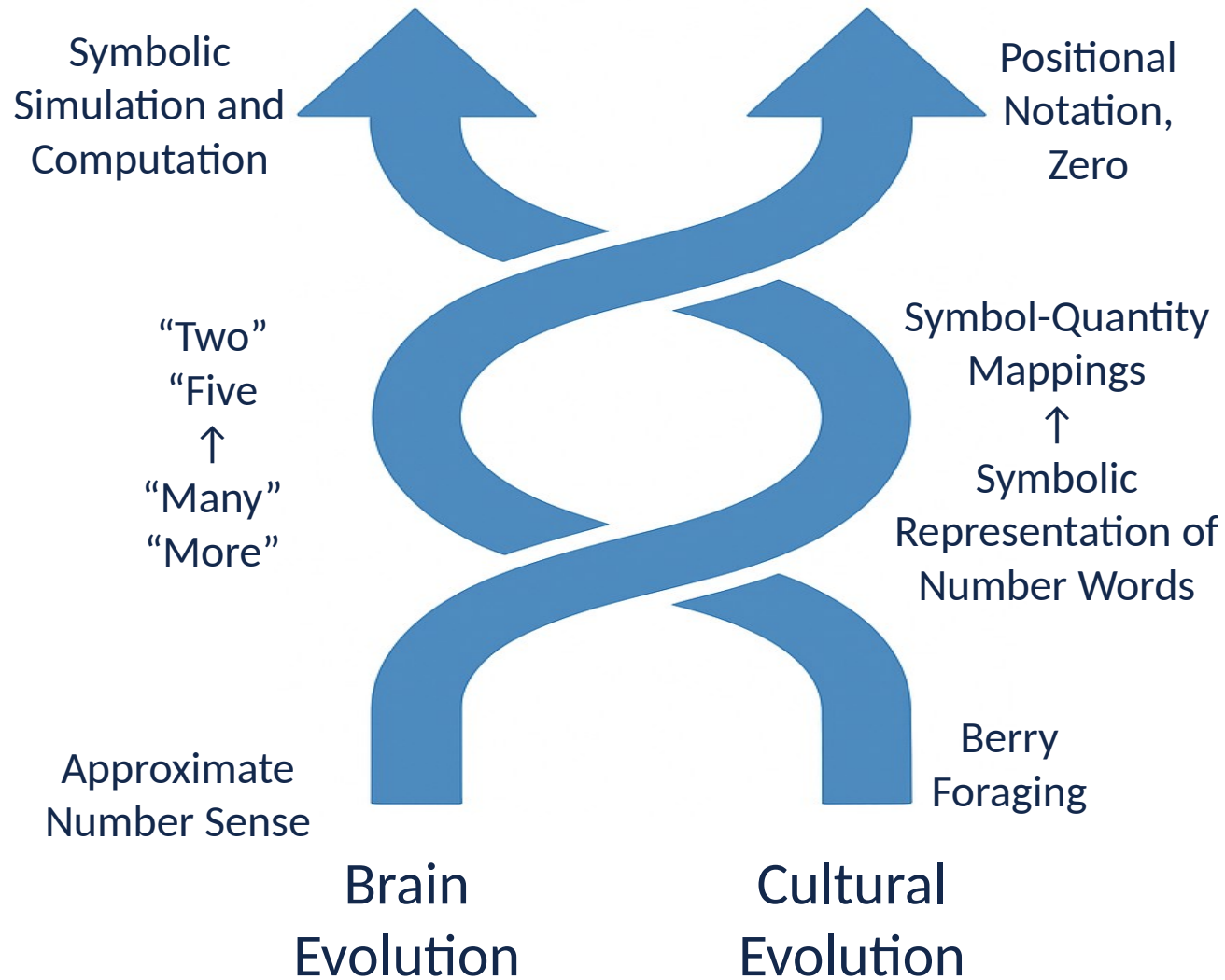


Cultural Evolution

- Ideas evolve too
- Variation, selection, inheritance
- A second system of evolution



Symbolic Cognition and the Cultural Explosion



Brain-Culture Coevolution

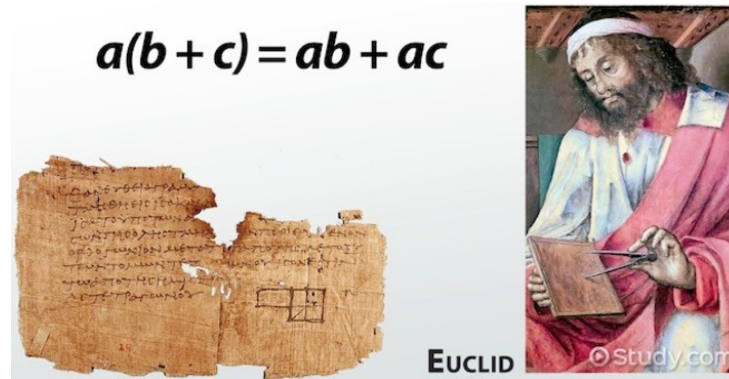
- Minds shape culture
- Culture reshapes minds
- A feedback loop across generations

Symbolic Cognition and the Cultural Explosion

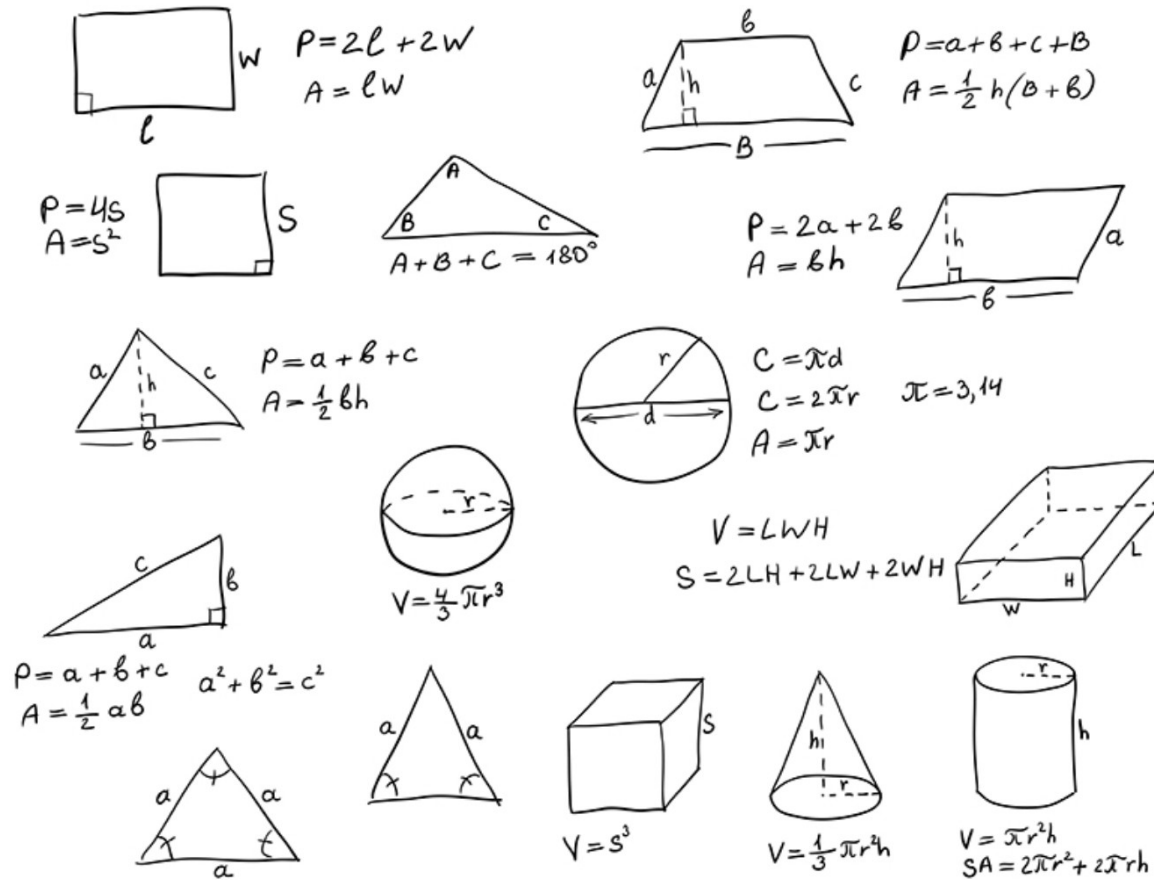


Externalizing Thought

- From mind to material
- Symbols as memory made visible
- Cognition moves into the world



Symbolic Cognition and the Cultural Explosion



Culture as Cognitive Technology

- Culture as mindware
- Tools for thinking, not just surviving
- Ideas that extend intelligence

Symbolic Cognition and the Cultural Explosion



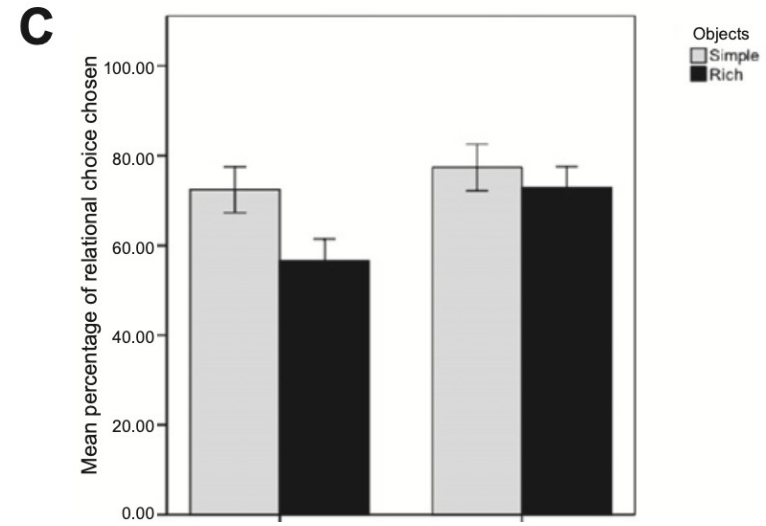
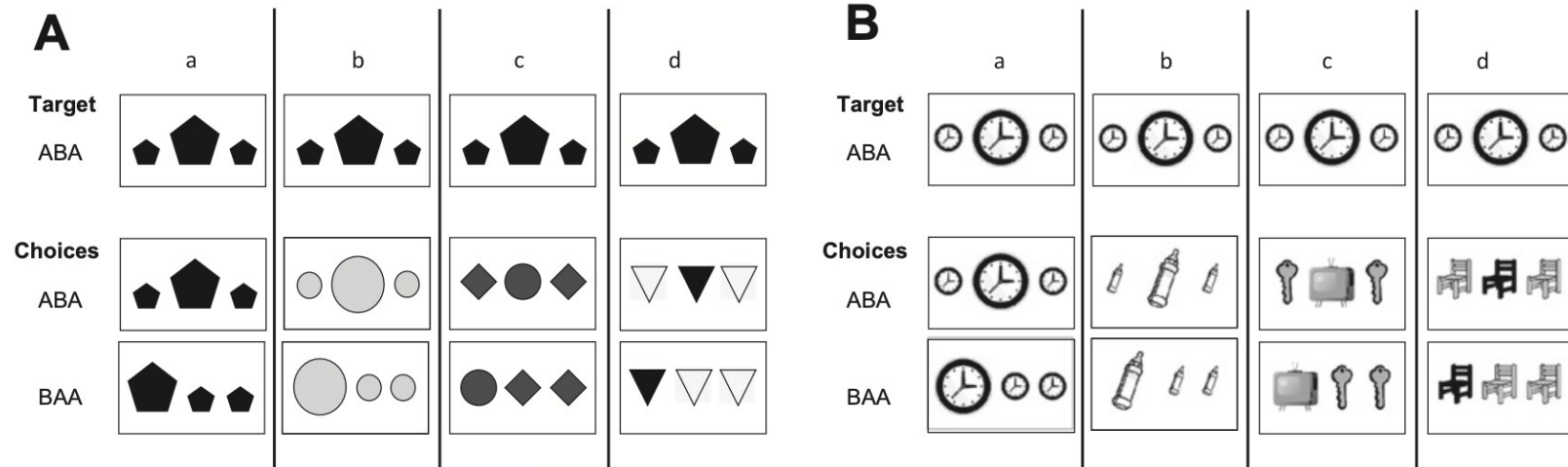
The Collective Brain

- No one mind holds it all
- Knowledge distributed across people
- Culture as a network of minds

Symbolic Cognition and the Cultural Explosion

Evidence from Cross-Cultural Cognition

- Different cultures, different minds
- Perception, reasoning, and self vary
- Culture tunes cognition



Symbolic Cognition and the Cultural Explosion

Symbols and Emotion

- Feeling together through symbols
- Ritual, art, and music as emotional code
- Emotion as the glue of culture



Celtic Triskellion



Centeōtl



Hongshan Dragon

The Irritating Gentleman
Berthold Woltze (1864)



Symbolic Cognition and the Cultural Explosion

There are lots of berries over the hill

Costs and Consequences

- Symbols empower—but also constrain
- Culture creates wisdom and bias
- The double edge of symbolic thought

$$\begin{array}{r} P \rightarrow Q \\ P \\ \hline \therefore Q \end{array} \qquad y = 3x + 5$$

If a feather is light, it doesn't weigh much

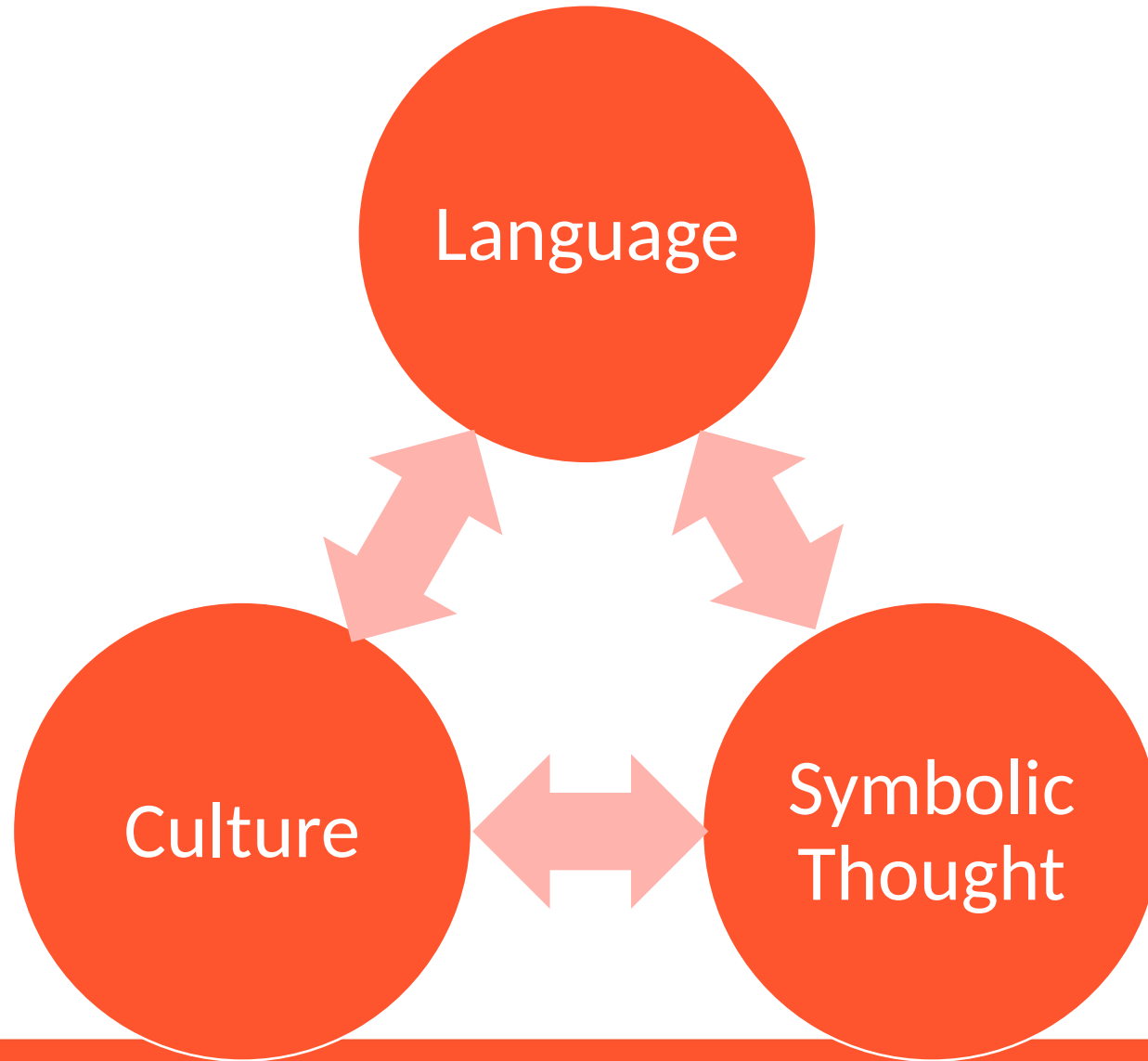
Love

Justice

Man
Woman

Us
Them

Symbolic Cognition and the Cultural Explosion



Integrating Levels: The Symbolic Feedback Loop

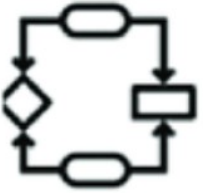
- Biology, cognition, and culture entwined
- Symbols linking brain and world
- An evolving loop of mind and meaning

Symbolic Cognition and the Cultural Explosion



Computation

What does the system do?



Algorithm

How does the system do it?



Implementation

How is the system realized?

Summary

- The rise of the symbolic species
- Mind, meaning, and culture coevolve
- From communication to civilization

Next Time

- Art, music, and storytelling
- Writing, literacy, and numerical cognition